Maharashtra

The Kharif season, stretching from June to October, is a crucial agricultural period for the state of Maharashtra, significantly influencing the livelihoods of millions of farmers. Characterized by the onset of the southwest monsoon, this season supports the cultivation of key crops such as rice, cotton, soybeans, pulses, and millets. However, climate variability, erratic rainfall, and changing weather patterns pose continuous challenges to sustainable agricultural production. In the light of these challenges, the Kharif 2025 Agro-Advisory aims to provide timely, region-specific, and science-based recommendations to empower farmers with the knowledge needed to make informed decisions.

REGIONAL ADVISORY

Maharashtra is divided into several agroclimatic zones, each with unique characteristics influencing crop selection, sowing windows, and input management. The following region-wise advisories aim to support farmers with tailored recommendations for improved productivity and resilience.

VIDARBHA REGION

The Vidarbha region of Maharashtra is composed of Bhandara, Chandrapur, Gondia Gadchiroli, Nagpur, Amravati, Akola, Yavatmal, Buldhana, Wardha, Washim and possesses state agricultural university viz., Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

Soil Testing & Preparation (All Crops)

Soil testing is crucial for effective fertilizer management, optimizing input use, and maintaining soil fertility. For proper soil sampling, the recommended depths are 15-20 cm for crops like sorghum, rice, groundnut, and wheat; 30 cm for cotton, sugarcane, and banana; and 30 cm for fruit trees, sampled from a distance of 1-1.5 feet from the trunk. Additionally, applying decomposers to crop residues like straw and leaves enhances organic matter content, improves soil structure, and boosts water retention, contributing to sustainable agricultural practices.

CEREALS CROPS

Paddy

Paddy after Transplanting

For optimal crop establishment, use certified seeds or ensure farm-saved seeds have over 70% germination. Perform salt treatment (300g salt/10L water), discard floaters, wash, dry, and treat with fungicide and Trichoderma (4-5g/kg seed). Sow from the 2nd week of June to 10th July (direct-seeded) or transplant from 2nd to 20th July. Deep plough with a sub-soiler, align with the slope, dig farm ponds, and prepare for drill sowing. Apply Pendimethalin 30% EC (50ml/10L water) after sowing and Bispyribac sodium 10% SC (6-7ml/10L water) 10-15 days later. Use 100:50:50 kg NPK/ha-full P & K with half N at planting; split the rest at tillering and panicle initiation. Begin seed preparation with a salt treatment (300 g salt per 10 L water); discard floaters, then wash, dry, and treat the seeds with fungicide and *Trichoderma* at 4-5 g/kg of seed. For optimal growth, direct seeding is recommended from the second week of June to July 10, while transplanting should be done between July 2nd to 20th July. Land preparation includes deep ploughing using a sub-soiler, ensuring cultivation is aligned parallel to the slope. Farm ponds should be dug for water harvesting, and the land should be prepared for drill sowing. Weed management involves applying Pendimethalin 30% EC (50 ml/10 L water) as a pre-emergence herbicide after sowing, followed by Bispyribac sodium 10% SC (6-7 ml/10 L water) as a post-emergence treatment 10-15 days later. Nutrient management consists of applying 100:50:50 kg NPK per hectare, with full doses of phosphorus and potassium and half of the nitrogen at planting. The remaining nitrogen is split and applied during the tillering and panicle initiation stages.

Maize

Fall armyworm attack in maize

Sowing of maize is recommended from mid-June to the first week of July, using the PDKV Arambha variety. Apply 20-25 cartloads of FYM per hectare before sowing. Treat seeds with biofertilizer at 25 g/kg. The recommended fertilizer dose is 120:60:60 (N:P:K), applied in three splits- 1/3 at sowing, 1/3 at 30 days after sowing (DAS), and 1/3 at 50 DAS. Maintain clean cultivation with deep ploughing, trap crops like Napier grass, bird perches, and pheromone traps. For pest management, spray 5% NSKE and release *Trichogramma* at 50,000 per acre weekly. For Fall Armyworm control, treat seeds with Cyantraniliprole + Thiamethoxam and apply Emamectin or NPV sprays if necessary.

COMMERCIAL CROPS

Cotton

High Density planting in cotton

The cotton crop is sown during May-June. Land preparation includes deep ploughing every three years for rainfed areas and annually for irrigated ones, followed by two harrowings and one pre-sowing harrowing. FYM or compost is applied at 12-15 cartloads for rainfed and 20-25 cartloads or 5 t/ha for irrigated fields, with 2.5 t/ha vermicompost if available. Seeds are treated with Imidacloprid or Thiomethoxam (7.5 g/kg), Thiram or Captan (3 g/kg), and biofertilizers *Azotobacter* (25 g/kg) and PSB (20 g/kg). Sowing times vary: 20-30 May for irrigated, early June for dry sowing, and 15-30 June post-monsoon. For Bt/Hybrid (irrigated), spacing is 120x90 cm with a seed rate of 2-2.5 kg/ha, while Deshi improved (rainfed) is sown at 60x15 or 60x30 cm with 9-15 kg/ha. Intercropping with green gram, black gram, or soybean (1:1) or with sorghum and pigeon pea (6:1:2:1) is recommended. Gap filling and thinning are done within 3-5 days after emergence. Irrigated Bt requires 40:60:60 NPK, while Hybrids need 35:50:50 NPK, both applied in two splits. Rainfed varieties receive 20-30:20-30:20-30 NPK as a basal dose using the ring method. Delayed application reduces yield by 10-40%. Foliar sprays of 2% urea (flowering) and 2% DAP (boll stage) are advised. Drip fertigation involves staged application: 10% at sowing, 20% at 20 DAS, 25% each at 40 and 60 DAS, and 20% at 80 DAS, with phosphorus applied as basal. Weed control is crucial for the first 60 days, requiring 3-4 hoeings and 2-3 weedings. Moisture conservation is done by opening furrows at 40-60 DAS. Rainfed cotton benefits from 2-3 protective irrigations during flowering and bolling.

Common pests like Jassid, Thrips, Aphids, and Mites are managed with Dimethoate 30 EC (10 ml/10 L) or Acetamiprid 20 SP (1.5 g/10 L), while whiteflies are controlled using yellow sticky traps (5 per acre) and Dimethoate 30 EC (33 ml/10 L). Bollworms are managed with Quinalphos, Spinosad, or Cypermethrin, and pink bollworm with pheromone traps, Neemark 300 ppm, deep ploughing, and heat treatment. Thirty yellow sticky traps and 2-3 light or pheromone traps per gin are suggested. Biopesticides like Neem 5%, Verticillium lecanii, and Beauveria bassiana are recommended in humid weather.

Plant Protection: Bacterial leaf blight is managed by treating seeds with Carboxin (1 g) and Thiram (3 g/kg), grey mildew with wettable sulphur (25 g/10 L) or Carbendazim (10 g/10 L), Myrothecium with Copper oxychloride (0.3%), Carbendazim (0.1%), or Mancozeb (0.25%), and Alternaria with seed treatment and Mancozeb (0.2%) sprays every 20 days. Para wilt is managed with a drench of 25 g Copper oxychloride and 150 g urea in 10 L of water.

Special recommendations include intercropping AKH-081 with green or black gram, spacing PKV DH-1 at 60x45 cm with 50:25:25 NPK (rainfed), and applying 2% urea (flowering) and 2% DAP (bolling) foliar sprays. For AKH-081 in shallow soils, 5 t/ha FYM and 25:12.5 N:P are advised. For pre-monsoon yield boost, 2.5 t FYM and 75:37.5:37.5 NPK/ha are applied.

Sugarcane

Sugarcane is planted during the Kharif season (June-July) with a duration of 12-15 months. It thrives best in well-drained deep loam soils with good water-holding capacity, avoiding saline or alkaline conditions. The climate should be warm and humid. Land preparation involves deep ploughing (1-2 times) followed by two cross harrowings, stubble and weed removal, and proper field leveling with drainage. Recommended varieties include CO-86032, CO-8014, CO-94012, VSI-434, and CO-M-0265, especially early-maturing ones. Healthy 9-10 month-old canes are selected for setts, taking the top two-thirds portion and avoiding flowering canes. The seed rate is 45,000-50,000 three-budded setts per hectare, approximately 6-8 tonnes. Before planting, setts are treated with 0.25% Bavistin or Carbendazim mixed with a 2.5% urea solution for 10-15 minutes.

Spacing varies: 90 cm for rainfed, 120 cm for irrigated or paired row trench planting, and 60 cm for High-Density Planting Systems (HDPS). Planting is done in furrows or trenches, with buds facing upward, covered lightly with soil. The fertilizer dose is 250:115:115 NPK kg/ha for irrigated crops and 200:90:90 NPK kg/ha for rainfed crops. At planting, 1/3 nitrogen (N) along with full phosphorus (P) and potassium (K) is applied, while the remaining N is split and applied at 45 and 90 days after planting (DAP). Fertilizers are placed in bands 8-10 cm deep and 10-15 cm away from the rows. Additionally, 25-30 t/ha of FYM or compost is applied before the final harrowing. Biofertilizers such as Azotobacter and PSB (2 kg/ha each) mixed with 100 kg of compost or FYM are added during planting. Intercropping with green gram, black gram, soybean, or groundnut is recommended during early stages. Light earthing up is done at 45-60 DAP, followed by full earthing up at 90-120 DAP. Trash mulching with cane trash after 60 DAP helps conserve moisture and control weeds.

For pest control, early shoot borer is managed by removing affected shoots, releasing *Trichogramma* (@ 50,000/ha/week), or applying Chlorpyrifos 20 EC (5 ml/L). Internode borer is controlled with Cartap hydrochloride 4G (10 kg/ha) or *Trichogramma* release. White grub management includes applying Phorate 10G (25 kg/ha) in moist soil. Red rot is prevented by planting resistant varieties, removing infected clumps, and treating setts with Carbendazim 0.1%. Smut is controlled by using disease-free setts, hot water treatment (50°C for 30 minutes), or fungicide dips. Wilt is managed through resistant varieties and crop rotation. For growth enhancement, 2% urea or 2% DAP is applied during tillering and grand growth stages if the crop appears pale or yellow. These practices ensure healthy growth and optimum yields.

PULSES

Soybean

Sowing of soybean is recommended between 20th June and 5th July, with a seed rate of 65 kg/ha and a spacing of 45 cm. Preferred varieties include MAUS-162, MAUS-612, PKV Amba, and PKV Yellow Gold. Seeds should be treated with a fungicide-insecticide combination like Azoxystrobin (10 ml/kg) and inoculated with Rhizobium and PSM (5 g/kg). For weed control, apply Pendimethalin 30% EC (3.25 L/ha) or Flumioxazin 50SC (250 ml/ha) as a pre-emergence herbicide, followed by

Imazethapyr 10% SL (1 L/ha) at 15-20 days after sowing (DAS) for post-emergence. Install sticky traps for whiteflies and monitor for hairy caterpillars, removing them manually if detected. Azadirachtin 1500 ppm (1 L/ha) is recommended as the first spray for *Spodoptera*, with additional insecticides as needed. For bacterial and fungal spots, apply fungicides at the first sign of symptoms. Nutrient application should follow the 25:60:40:20 NPKS kg/ha guideline, using 56 kg urea, 375 kg SSP, and 67 kg MOP. For stem fly, wilt, root rot, and fungal diseases, treat seeds with Azoxystrobin 2.5%, Thiophanate 11.25%, and Thiamethoxam 25% FS (10 ml/kg of seed). Maintain weed-free fields during the early growth stage, destroy host plants on bunds, and remove virus-infected plants promptly to prevent disease spread. Use 30-40 yellow sticky traps per acre to control whiteflies and sap-sucking pests. Apply balanced fertilizers to prevent excessive nitrogen, which attracts sap-sucking pests and promotes viral spread. For pest management, spray Imidacloprid in 10 L of water as required. To control whiteflies that transmit yellow viral disease, use a combination of Acetamiprid 25% + Bifenthrin 25% WG (5 ml/10 L water) or any recommended insecticide as per label instructions.

Pigeon Pea

In Maharashtra, recommended pigeon pea varieties include BDN-716, BSMR-736, PKV Tara, PKV Ashlesha, Vipula, Asha, and Vaishali (BSMR-853). Early varieties are sown in the first fortnight of June, while medium and late varieties are sown in the second fortnight. The seed rate for early varieties is 20-25 kg/ha with a row spacing of 45-60 cm, whereas medium and late varieties require 15-20 kg/ha with 60-75 cm spacing. For seed treatment, Thiram + Carbendazim or *Trichoderma viride* combined with *Rhizobium* and PSB culture is recommended. Alternatively, Carboxin + Thiram (3 g/kg) and *Trichoderma* (4 g/kg) can be used. Fertilizer application includes 25-30 kg N, 40-50 kg P_2O_5, and 30 kg K_2O per hectare as a basal dose, along with 20 kg of sulfur and 3 kg of zinc if deficiency is observed. Weed management involves the pre-emergence application of Pendimethalin at 0.75-1 kg a.i./ha, followed by two hand weedings at 20-25 and 45-50 days after sowing (DAS). Irrigation should be provided during critical growth stages such as branching, flowering, and podding if dry spells occur.

To manage wilt, seeds are treated with *Trichoderma viride* at 10 g/kg, with additional soil application. For Sterility Mosaic Disease (SMD), spray Fenazaquin (1 ml/L) at 45 and 60 DAS. Pod borers are controlled using Emamectin benzoate at 220 g/ha and pheromone traps, while pod fly infestations are managed with a 5% spray of NSKE at 50% flowering. Proper crop management ensures healthy growth and higher yield.

Mung Bean

Sowing should be done from the last week of June to the first week of July, following the onset of monsoon. To manage leaf crinkle, treat seeds with Thiamethoxam 70WS (5 g/kg) and spray Thiamethoxam 25WG (4 g/10 L) at 21 and 35 days after sowing (DAS). Recommended varieties include BM-4. Apply a 40 ppm Chitosan spray at 25 and 40 DAS. For Cercospora management, spray Metiram + Pyraclostrobin (0.3%) at the first sign of infection and repeat after 15 days.

Black Gram

Sowing is recommended from mid-June to early July with the onset of monsoon. Preferred varieties include TAU-1 and T9. Bio-prime the seeds with Rhizobium and PSB (25 ml/kg) for 12 hours before sowing. Apply 10:20 kg N:P_2O_5 per hectare, achieving a 50% saving in nitrogen and phosphorus. For pest management, spray Monocrotophos (12.5 ml/10 L) at the bud stage, followed by Chlorantraniliprole (2.5 ml/10 L) after 15 days.

FRUIT CROPS

Citrus (Mandarin)

Mandarin requires well-drained light to medium loam soil with a pH of 6.5-7.5, avoiding saline or alkaline conditions. It is planted during June-July (monsoon) or January-February (spring) with a spacing of 6 m × 6 m (277 plants/ha) or 5 m × 5 m in lighter soils. Pits (1 m^3) are prepared with topsoil, 20-25 kg FYM, 1 kg SSP, and 500 g Neem cake before planting. Recommended varieties include Nagpur Mandarin, PKV Shanti, PKV Nira, and PKV Garima. Young plants need light irrigation every 7-10 days, while bearing trees require irrigation at critical stages-flowering, fruit set, and fruit development-avoiding over-irrigation during flowering and monsoon. Fertilizer application starts with 100 g N, 50 g P, and 50 g K in the first year, gradually increasing to 500 g N, 250 g P, and 250 g K from the fifth year onwards, split between June-July and September-October. Apply 10-30 kg FYM per tree annually in June. Intercropping with cowpea, green gram, black gram, or marigold is recommended during the early years. For pest control, spray Dimethoate or Imidacloprid for citrus psylla and leaf miner; Neem oil or Acetamiprid for black fly and whitefly; poison bait or bagging for fruit-sucking moth; and Chlorpyrifos or kerosene for trunk borer. Canker is managed with a 1% Bordeaux mixture or Streptocycline (100 ppm) + Copper oxychloride (0.2%). For gummosis, scrape bark and apply Metalaxyl paste while ensuring good drainage. Greening disease is prevented by using disease-free budwood and controlling the psylla vector.

SPICES

Turmeric

Rhizome planting time: Mid-May to Mid-June (preferably with onset of monsoon or just before). Varieties: Suguna, Suvarna, Prabha. The land is prepared with the receipt of early monsoon showers. The soil is brought to a fine tilth by giving about four deep ploughings. Hydrated lime @ 500 kg/ha has to be applied for laterite soils and thoroughly ploughed. Immediately with the receipt of pre-monsoon showers, beds of 1.0 m width, 15 cm height and of convenient length are prepared with spacing of 50 cm between beds. Planting is also done by forming ridges and furrows. Treat rhizomes with a fungicide solution (like Mancozeb or Carbendazim) before sowing to protect against early diseases. You can also use bio-fungicides like *Trichoderma*.

Plant rhizomes at appropriate spacing (typically 20-30 cm between plants and 45-60 cm between rows). Ensure adequate moisture at the time of sowing. Depending on rainfall, you might need to provide initial irrigation.

FLD on Turmeric

Pest, Nematode, soil born pathogen: Biological control - At the time of planting, apply 25 g powdered neem cake and mix well with the soil in each pit taken at a spacing of 20-25 cm within and between rows or application of neem cake @2 t/acre is also desirable. Plan for timely weed control through manual weeding or appropriate herbicides as per recommendations.

LIVESTOCK MANAGEMENT

In Livestock shed use of 0.01 % Deltamethrin spraying for control of ecto and endo parasites 15 days intervals. Provide ample amount of clean and cold water to livestock to avoid Colitis disease. Use of white paint lime to drinking water tank. Vaccination of livestock should be done twice in a year against FMD. Livestock deworming should be done once in a six month. In case of buffalo wallowing should be done in summer season for enhancing the milk yield and to maintain the body temperature. Grooming of milch animal to avoiding the falling of loose hairs in the milk and to get the clean milk production and enhancing the blood circulation. Washing of animal specially in summer season remove the dirt and dust and maintain the body temperature and to avoid the heat strock. As per thumb rule provide animal feed should be 60 % green fodder and 40% dry fodder. In case of milch animal give 30 gm mineral mixture in case of cattle and 60 to 80 g in case of Buffalo should be given in the diet.

ENGINEERING

Use sub-soiler for deep ploughing. Cultivate parallel to slope. Tools used are Paddy transplanter, power weeder, Cono weeder. Size of farm pond: 20x20x3 m per 2 ha.

MARATHWADA REGION

Marathwada region of Maharashtra is comprised of Aurangabad, Jalna, Beed, Latur, Osmanabad, Hingoli, Parbhani, Nanded. The region has moderate rainfall (600-800 mm) and has medium to deep black soils. This region possesses state agricultural university viz., Dr. Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani.

CEREALS CROPS

Maize

Maize should be sown at 75 x 20 cm or 60 x20 cm as per soil type by dibbling method with first onset of monsoon after receiving 75-100 mm rainfall. Improved and high yielding notified varieties or research hybrids should be used for sowing for achieving higher yield. Recommended dose of 100:75:75 kg NPK should be applied to maize out of which 50 % N and full P and K should be given at the time of

Maize observation in field

sowing while remaining 50 % N should be applied after 30 DAS. For the control of Fall Army Worm, treat the seed with 3 g/ Thiram per kg of seed, spray 5 % NSKE withing 30 days of sowing or at 5 % infestation. In addition, spray Emamactin Benzoate 5 % @ 4 g/ 10 L of water at 10-20 % infested plants. Earthing up should be done during dry spell for in situ moisture conservation.

COMMERCIAL CROPS

Cotton

Whitefly on cotton

Cotton should be sown after onset of monsoons or after 1st June only for effective management of pink bollworm. Well decomposed FYM should be applied to cotton @ 10 ton/ha before sowing. Recommended fertilizer dose of 120:60:60 kg NPK/ha should be applied to rainfed cotton out of which 40% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 30 % N should be applied after 30 DAS and 30 % after 60 DAS. Use of high yielding, short duration cotton variety like CICR-14, CICR-19, PKV-081, Suraj BT, Rajat BT, NH-1901,1902,1903,1904 etc. For irrigated cotton apply 150:75:75 kg NPK/ha out of which 20% N & complete P & K should be applied as a basal dose at the time of sowing. Remaining 40 % N should be after 40 DAS and 30 % after 60 DAS. Closer spacing and High-Density Cotton Planting (HDPS) should be adopted on medium and light soils respectively for higher yields by keeping plating distance of 90 x 30 cm for medium soils and 90 x 15 cm for light soils which will accommodate approximately 14000 and 29000 plants per acre respectively. Canopy management should be done by cutting monopodia at 45-50 DAS and nipping the tip at 90 DAS supported with one or two sprays of growth retardants if required under expert guidance. Earthing up should be done in cotton during dry spell in Kharif season to reduce evaporation losses of soil moisture. Follow IPM package for the control of pests and diseases in cotton as given below. Select short duration and sucking pest tolerant variety for sowing

a. Follow crop rotation avoid cotton sowing followed by tomato, chickpea, etc.

b. Sowing of intercrop or mixed crop like maize, cowpea, Black gram, green gram and use of trap crop like marigold, castor, etc.

c. Sowing of okra as a trap crop in cotton for the control of Pink Bollworm.

d. Avoid plant growth promoters and pesticides during first three months and use botanical pesticides or biopesticides

e. Spray of 5 % NSKE at 60 DAS

f. Install 40 pheromone traps per ha for mass trapping of pink bollworm.

PULSE CROPS

Soybean

The own seed of Soybean should be used for next Kharif season after testing its germination to save money as well as it will help to face the situation of seed shortage in the market. Seed germination should be tested before sowing and good quality seed with more than 70 % germination should be used for sowing only. Improved varieties like MAUS-162, MAUS-612, MAUS-725, Phule Durva, Phule Kimaya, PDKV Yellow Gold, PDKV Amba should be used for sowing. Soybean seed should be treated with Thiram @ 3 g/kg or *Trichoderma* @ 4 g/kg of seed and inoculated with *Rhizobium* and PSB @ 100 ml/10 kg of seed before sowing. Sowing of soybean should be avoided in the fields where soybean seed production was taken in the last summer season. Follow IPM package for the control of pests and diseases in soybean as give below:

a. Sowing should be completed before 15th July to avoid pest infestation.

b. Seed treatment of Thiram @2-3 g per kg or *Trichoderma* @5 g per kg of seed.

c. Seed treatment with Thiamethoxam 30 % @ 10 ml per kg of seed for management of stem fly.

d. Use border crop of castor and sunflower as a trap crop for *Spodoptera*.

e. Install 16 to 20 bird perches per ha.

f. For management of Spodoptera spray 5 % NSKE or *Beauveria bassiana* or *Numeria releyi* @ 40 g or *Bacillus thuringiesis* @ 20 g per 10 L of water.

Pigeon Pea

Wilt in Pigeon pea

The Sowing time for early varieties is first fortnight of June and medium/late: Second fortnight of June to first fortnight of July. Seed rate for early: 20-25kg/ ha spacing: 45X60cm R-R and Medium to Late: 15-20kg/ ha spacing: 60-75cm R-R. Use wilt and sterility mosaic resistant medium to early duration varieties like BDN-711, BDN-716, BDN-2013-41 (Godavari), PKV Tara, Phule Rajeswari, etc to ensure higher yield in adverse climatic conditions. Treat the seeds with biofertilizers and apply recommended doses of fertilizers as per recommendations of concerned universities Ensure arrangement of one or two protective irrigation wherever possible. Intercropping of Pigeon pea + Soybean with 1:2 or 2:4 ratio should be taken for higher yield and monitory returns. Install 5 pheromone traps per hectare for monitoring and spray 5 % NSKE at flowering stage for the control of *Helicoverpa*.

Green Gram and Black Gram

Sowing of green gram should be completed with first onset at minimum 80 mm of rainfall. Improved varieties of green gram like BPMR-145, BM-2003-02, Phule Chetak, etc and AKU-10-1 (Black Gold), AKU-15, etc of black gram should be used for

sowing. Seed of green gram and black gram should be treated with Thiram @ 2 g/ kg followed by seed inoculation of Rhizobium and PSB biofertilizers @ 10 ml / kg of seed. Green gram and black gram can be taken as intercrop in cotton at 1:1 ratio for higher yield and monitory returns.

FODDER CROPS

Sorghum

Improved varieties of Sorghum like CSV-20, CSV-23, CSV-27, PVK-801, PVK-809, CSH-25 (Parbhani Sainath), biofortified variety like Parbhani Shakti should be used for sowing. Seed should be treated with Thiram @ 3 g/ kg followed by seed inoculation of *Azotobacter* and PSB biofertilizers @ 10 ml / kg of seed. Kharif Sorghum should be sown at 45 x 12.5 cm spacing by using 7.5 kg seed rate per hectare. Recommended fertilizer dose of 80:40:40 kg NPK should be applied to Kharif Sorghum out of which 50 % N and full P and K should be given at the time of sowing while remaining 50 % N should be applied after 30 DAS. For the control of Stem Borer and Fall Army Worm, spray 5 % NSKE or Quninolphos 35 % EC @ 15 ml / 10 L of water within one week from sowing. Take another spray at the interval of 10 days if required.

Pearl Millet

Improved varieties like Shraddha, Saburi, AIMP-92901 and biofortified varieties like AHB-1200 Fe, AHB-1269 Fe should be used for higher yield. Recommended fertilizer dose of 60:30:30 kg NPK should be applied to Pearl Millet out of which 50 % N and full P and K should be given at the time of sowing while remaining 50 % N should be applied after 30 DAS. Seed should be treated with Thiram @ 3 g / kg of seed followed by Metalaxil 35 SD @ 5 g / kg of seed and then inoculate with *Azotobacter* and PSB biofertilizers @ 25 g / kg of seed.

COMMERCIAL CROPS

Sugarcane

Deep ploughing followed by harrowing and levelling; ensure proper drainage as the region is prone to waterlogging. Use high-yielding, disease-resistant varieties like Co-86032, Co-94012, Co-8014, Co-M-0265 (Phule-265), or Co-C-671. Ideal time for adsali (pre-kharif) planting is July-August; Kharif planting should be done in June-July. Use 3-budded setts from healthy, disease-free canes; dip in fungicide solution (e.g., Bavistin @1g/liter) before planting. Adopt 90 cm row-to-row and 30 cm plant-to-plant spacing for proper growth and aeration. Apply 250:115:115 kg NPK/ha in three splits (at planting, 45-60 DAS, and 90 DAS). Use organic manure or FYM @ 25-30 t/ha for better soil health. First irrigation immediately after planting, then every 8-10 days depending on rainfall and soil moisture. Perform 2-3 hand weeding/inter culturing at 30, 60, and 90 DAS. Use pre-emergence herbicides like Atrazine @ 1.5 kg/ha.

Pest & Disease Management:

Early shoot borer: Remove and destroy infested shoots; apply Chlorantraniliprole or Carbofuran granules in affected furrows.

Top shoot borer/ White grub: Apply *Trichoderma* or *Beauveria bassiana* in soil and spray Chlorpyrifos if needed.

Red rot: Use resistant varieties and avoid ratooning affected crops. First earthing up at 45-60 DAS; final at 90-120 DAS to provide support and prevent lodging. Mulch inter-row spaces with cane trash to conserve moisture and suppress weeds. Grow short-duration legumes (moong, urad) in early stage for additional income and nitrogen fixation.

FRUIT & VEGETABLE CROPS

Citrus

For new plantation of citrus (Sweet Orange and Lemon), well-drained soil with less than 10 % free lime should be selected. Use quality and disease-free planting material from registered nurseries for new plantation. Sweet Orange crop should be planted at 6 x 6 m spacing accommodating 277 plants per hectare. Only one bahar either Ambiya or Mrig should be taken based on water availability for higher yield and more life of sweet orange orchard. Irrigation management should be done in Ambia Bahar to citrus orchard. Take a spray of potassium nitrate @ 15 gm per litre of water. Application of Bordo paste (1:1:10) should be done on tree trunk before onset of monsoon to protect the crop from gummosis and dieback. Application of Trichoderma viride, Pseudomonas fluorescence, Glomas mycorrhiza @ 100 g each in 1 kg compost with 200 g Ferrous Sulphate and Zinc Sulphate to each gummosis affected plant. For management of fruit sucking moth, adopt Integrated Pest Management practice which include, destruction of weed host, applying smoke, collection and destruction of fallen fruits, use of light trap, use of poison baits, covering tree with nylon net of 0.5 mesh and spraying of Horticulture Mineral Oil @ 500 ml/ ha, etc.

Mango

Preferably Marathwada Keshar variety should be selected for plating in Marathwada region. Organic mulching should be done and the orchard should be irrigated during early in the morning, at evening or during night hours. To protect the mango fruits from the incidence of fruit fly, install "Rakshak fruit fly trap" developed by university @ 4 traps per hectare. Fallen fruits should be collected and destroyed to keep orchard clean. Fruit fly lays its eggs in the fruits below outer cover of the fruit when the fruit is about to mature. The larvae develop on the pulp of the fruit. Hence, the infested fruits should be destroyed. The soil should be ploughed below the tree. After harvesting of mango, one spray of Hexaconazole fungicide should be taken @ 10 ml/10 L of water for prevention of blight.

Advisory services for Mango fruit orchard visit

Banana

Sword suckers of 3-4 months age with 450 to 750 g should be used for plantation of Banana. For protection of banana orchard from hot air, make use of green net in southwest direction of orchard. Earthing up should be done in banana orchard. Provide shade to newly planted or small plants to protect them from high temperature. Organic mulching should be done in banana orchard. Irrigate banana orchard during early in the morning, at evening or during night time.

Grapes

Management of April pruning in grape orchard should be done as per availability of water. During rest period, provide only need based irrigation to protect the existing leaves from drying, and also contribute towards increasing the reserves of the vines through photosynthetic activity. Care should be taken to reduce/stop the water in case new growth is observed on the shoot. If rainfall exceeds 2.5 mm, no irrigation should be applied especially during rest period.

Custard Apple

Prefer improved varieties like Balanagar, Red Sitaphal, Arka Sahan, and NMK-1 for better yield and fruit quality. Kharif planting should be done in June-July, with the onset of monsoon. Adopt spacing of 5 m × 5 m or 6 m × 6 m (approx. 270-400 plants/ha). Dig pits of 60×60×60 cm and fill with a mixture of topsoil + 10-15 kg FYM + 1 kg neem cake + 100 g SSP.

Fertilizer Management (per plant/year): For fertilizer management, during the first year, apply 50 g of nitrogen (N), 25 g of phosphorus (P), and 25 g of potassium (K) per plant. For mature plants, use 500 g of N, 250 g of P, and 250 g of K, along with 20 kg of farmyard manure (FYM) per plant annually. Apply the basal dose of fertilizers in June-July and split the nitrogen dose by applying the second half after fruit set.

Pest and Disease Management:

Fruit borer: Collect and destroy infested fruits. Spray Spinosad @ 1 ml/L . Mealybug: Use neem-based sprays or release natural predators like ladybird beetles.

Leaf roller: Spray Dichlorvos or Quinalphos as needed. Anthracnose (fruit rot): Spray Copper oxychloride (3g/l) or Mancozeb at 15-day intervals. Avoid overcrowding and prune to improve air circulation.

Pomegranate

Ideal planting during June-July (Kharif) with the onset of monsoon for rainfed or semi-irrigated conditions. Varities recommended are Bhagwa, Ganesh, Arakta, Mridula, and Super Bhagwa for high yield and export quality. Spacing & Planting: Spacing of 5 m × 5 m (400 plants/ha). Use healthy, disease-free, 1-year-old saplings. Prepare pits of 60×60×60 cm and fill with topsoil + 15-20 kg FYM + 1 kg neem cake + 1 kg super phosphate. Although drought-tolerant, irrigation at 10-15-day intervals after monsoon ensures better flowering and fruit setting. Use drip irrigation to conserve water and ensure uniform moisture.

Fertilizer Management (per plant/year): In the first year, apply 250 g of nitrogen (N), 125 g of phosphorus pentoxide (P_2O_5), and 125 g of potassium oxide (K_2O) per plant. Starting from the third year, during the bearing stage, increase the fertilizer application to 625 g of N, 250 g of P_2O_5, 250 g of K_2O, along with 20-30 kg of farmyard manure (FYM) per plant. Apply the fertilizers in split doses, with the first application in June-July and the second after fruit set.

Pest and Disease Management:

Fruit borer & Thrips: Spray Spinosad @ 1 ml/l or Imidacloprid @ 0.3 ml/l.

Stem borer: Plug holes and apply insecticide paste.

Aphids & Mealybugs: Spray neem oil (3%) or Dimethoate @ 1.5 ml/l.

Bacterial blight: Apply Bordeaux mixture (1%) or Streptocycline + Copper oxychloride.

Papaya

Drenching of copper oxychloride 2 gm/litre or Bavistin 1gm/litre of water to carry out if newly emerged papaya seedlings affected by root rot /Pythium fungal disease.

Chilli

To avoid leaf curl infestation in chilli transplanting of seedlings should be done before 30th June. Pest resistant or tolerant varieties like like Pusa Jwala or Pusa Jyoti should be used for planting.

Tomato

Triple disease resistant varieties of tomato like Arka Rakshak or Arka Samrat should be used for higher yield.

Okra

For management of fruit borer in okra take a spray of 5 % NSKE or Chlorantraniliprole 18.5 % SC @ 2.5 ml or Cypermethrin 10 % EC @ 15 ml or Quinalphos 25 % EC @ 20 ml per 10 L of water. Harvesting of mature vegetables should be done early in the morning. Irrigation management should be done early in the morning, at evening or during night time in vegetable crops as per availability and requirement of crop. Management practices should be followed to protect the vegetable crops from hot air.

Brinjal

If incidence of shoot and fruit borer observed in Brinjal, collect and destroy caterpillar or Spray Neemark 4 % or Cypermethrin 25% EC 5 ml or Chlorpyriphos 20 EC 20 ml or Deltamethrin 1 % or Traizophos 20 ml per 10 litres of water.

Turmeric

Harvesting, boiling, drying, polishing operations are being carried out in turmeric crop. To save the crop from unseasonal rains or hailstorm, crop should not be stored in open place. The harvested turmeric crop should be covered by plastic paper so that it should not be damaged by rains.

Animal Protection

To protect the animals from high temperature, apply white paint to the roof of shade. Apply mulching on the roof of shade with the help of dry grass and trash of sugarcane which reduces the temperature in shade. If water is spry on mulches or use of foggers will cool down the temperature of shade. Precaution should be taken so that the harvested dry Stover of sorghum may not damage due to rainfall, if it happens the quality as well as storage capacity of dry Stover decrease and animals not prefer such type of fodder.

Protection with gunny bags

Maize

In the Kharif season, the ideal sowing time is from June 15 to July 15, depending on the onset of monsoon rains. Apply 120 kg Nitrogen (N), 60 kg Phosphorus (P_2O_5), and 40 kg Potassium (K_2O) per hectare. At the time of sowing, apply half the nitrogen and the full dose of phosphorus and potassium. Apply the remaining nitrogen in two equal splits: one at 30 days and the other at 50 days after sowing. In case of micronutrient deficiencies, apply 5 kg/ha of Zinc Sulphate ($ZnSO_4$) and 5 kg/ha of Ferrous Sulphate ($FeSO_4$) at sowing and again 30 days after sowing. The first weeding should be done 20 to 25 days after sowing, and the second weeding around 45 days after sowing. To control weeds chemically, apply Atrazine 50% WP @ 1 kg/ha as a pre-emergence herbicide within 2-3 days of sowing. It should be applied uniformly on moist soil using a knapsack sprayer.

COMMERCIAL CROPS

Sugarcane

Well-drained deep loam with good water holding; avoid saline/alkaline soils; requires warm humid climate. Deep ploughing (1-2 times) + 2 cross harrowing; remove stubbles & weeds; level field with proper drainage. Varities like CO-86032, CO-8014, CO-94012, VSI-434, CO-M-0265 (early maturing recommended). Use 9-10-month-old healthy canes; take top 2/3rd portion; avoid flowering canes. Recommended seed rate is 45,000-50,000 three-budded setts/ha (~6-8 tonnes/ha). Treat with 0.25% Bavistin or Carbendazim + 2.5% urea solution for 10-15 minutes before planting. Avoid the use of infested Sets/seedlings. Paired Row planting system provides easy access for plant protection application. Judicious use of nitrogenous fertilizers to prevent infestation of sucking pests like Wooly aphids, Pyrilla, Scale insect, whitefly, etc.

Plant Protection: Remove and destroy the lower 4-5 leafs of cane to prevent infestation of Pyrilla and Scale insect. Use pheromone traps for the monitoring of Early Shoot Borer, Internode Borer and Top Borer and fortnight release of *Trichogramma* parasitoid insect (Trichocard 1.50 eggs/ha) to check the all-stem borer infestation. Preventive management of White Grub should be done by installing light traps in the field (1 trap/acre), collection and destroying beetles from bund trees during evening time. To control the initial infestation of white grub, apply *Metarhizium anisopliae* 15 kg or *Heterorhabditis indica* 7 kg per hectare as a biological insecticide in the month of June and repeat the application one month after first when incidence level is high. If the above treatments found unable to control follow the application of a mixed insecticide of Fipronil 40% + Imidacloprid 40% e at the rate of 15 ml per 10 litres of water. Also, if there is a high incidence of this pest, spray with Chlortraniliprole 18.50 SC @ 3-4 ml per 10 L of water. If symptoms of Rust disease observed, spray crop with Mancozeb 0.3% (30 g in 10 L of water) or Tebuconazole 0.1% (30 g in 10 litres of water). Regular monitoring of woolly aphids should be done, if colonies are found, they should be removed and destroyed, and predatory insects like *Conobathra, Micromus, Depha aphidivora* etc. should be conserved. For the Management of Smut disease spray the crop with Carbendazim 50 WP @ 10 g in 10 litre water. For the management of

Pokkah boing Disease spray the crop with Mancozeb 75 WP @ 30 g or Carbendazim 50 WP @ 10 g in 10 litre of water. For the management of Rust and Leaf spot diseases spray the crop with Mancozeb 75 WP @ 30 g or Tebuconazole 25 EC @ 10 ml in 10 litre water.

PULSE CROPS

Pigeon pea

Select proper sowing windows based on maturity groups. Early varieties can be sown in First fortnight of June and Medium/Late varieties in Second fortnight of June. Ensure flowering avoids low sunlight/cloudy weather to prevent poor pod formation. Use well-drained black cotton soil (pH 7.0-8.5). Adopt deep ploughing or BBF/Ridge & Furrow systems to improve drainage and aeration. Raised bed planting and appropriate row spacing (e.g., 4-5 ft) for better management and intercropping potential. Use quality seeds with appropriate seeding rates: Early: 20-25 kg/ha. Medium/Late: 15-20 kg/ha. Treat seeds with Trichoderma viride, Thiram + Carbendazim, and Rhizobium + PSB for disease prevention and better nitrogen fixation. Fertilization: Apply 25-30 kg N, 40-50 kg P_2O_5, 30 kg K_2O/ha at sowing. Incorporate Sulphur, Zinc, and Iron if soils are deficient. Follow soil test-based recommendations for precision nutrition. Irrigation & Drainage: Though drought-tolerant, apply 3 key irrigations during branching, flowering, and podding stages. Use ridge planting in poorly drained soils to avoid waterlogging. For weed Management: Pendimethalin (0.75-1 kg a.i./ha) as pre-emergence to suppress early weeds. Conduct two mechanical weedings before flowering (20-25 DAS & 45-50 DAS). Cropping and Intercropping Systems: Utilize inter-row space with short-duration crops: mung, urd, cowpea, , and recommended intercropping arwe Pigeon pea-Urd-Wheat, Pigeon pea-Sugarcane, Pigeon pea (early)-Potato-Urd bean.

Pest & Disease Management: Use resistant varieties (e.g., Asha, BSMR-853, JKM-189). Follow integrated pest management (IPM): pheromone traps, Neem products, timely chemical sprays (Emamectin, Indoxacarb).

Soybean

Soybean should be sown by the row-to-row spacing is 45 cm, and plant-to-plant spacing is 7 to 10 cm. A seed rate of 65 to 75 kg per hectare is sufficient, depending on the seed size and germination percentage.

Soybean should be treated with Thiram @ 3g/kg or *Tricoderma* @ 4 g/kg of seed inoculum like *Rhizobium* and PSB @ 100ml/10kg of seed before sowing. For higher production use new released high yielding varieties like Phule Sangam (KDS 726), Phule Kimaya (KDS 753), Phule Durva (KDS 992. Apply NPK @ 50:75:40 Kg/ha at the time of sowing. Use Broad bed furrow planter (BBF) for sowing. It will be beneficial during less rainfall situation and excessive rainfall. For chemical weed control, Pendimethalin at 1.0 kg active ingredient per hectare can be applied as a pre-emergence herbicide immediately after sowing. Imazethapyr at 75 grams active ingredient per hectare can be applied as a post-emergence herbicide at 15-20 days after sowing to control broadleaf and grassy weeds.

FRUIT CROPS

Pomegranate

Fruit cover in Pomegranate

For new plantations dug the pits at recommended planting distance. Procure seedlings or grafts from licensed fruit crop nursery. Pit filling may be completed before arrival of monsoon rains. Alternatively raised beds of 3 feet width and 1 feet height may be prepared at 15 feet apart. After two three showers of heavy rains go for transplanting of plants. High yielding and market acceptable varieties like Bhagwa, Super Bhagwa, Solapur Red are recommended. Soil application of bio-pesticides like *Pseudomonas* and *Trichoderma* is suggested as preventive measure against *Pythium, Phytophthora* and *Fusarium*. Inter crops of seasonal vegetables and flowers may be followed. In existing orchards remove water suckers. Foliar spray of Potassium Nitrate @5 g per liter of water is recommended. Ensure proper drainage of water. Spray *Verticellium lecanii* @ 40 gm per 10 L water for the control of mealy bugs.

Management of fruit spot/ rot problem spray Carbendazim 12% + Mancozeb 63% WP @ 500 gm in 200 litres of water during second week of June. During first week of July spray Copper oxychloride 400 gm +Streptocycline 20 gm in 200 litres of water for the management of bacterial spot. Spray Spinosad 0.4 ml per L of water or lambda cyhalothrin 10WP 1.25g per L of water during first week of May for the management of pomegranate butterfly.

Custard Apple

For new plantations dug the pits at recommended planting distance. Procure seedlings or grafts from licensed fruit crop nursery. Pit filling may be completed before arrival of monsoon rains. After two three showers of heavy rains go for transplanting of plants. High yielding and market acceptable like varieties Balanagar, Dharur, Daulatabad Selection are recommended. In existing orchards ensure proper drainage of water. Soil application of bio-pesticides like Pseudomonas and *Trichoderma* is suggested as preventive measure against Pythium, Phytopthora and Fusarium.

Guava

For new plantations dug the pits at recommended planting distance. Procure seedlings or grafts from licensed fruit crop nursery. Pit filling may be completed before arrival of monsoon rains. Alternatively raised beds of 3 feet width and 1 feet height may be prepared at 15 feet apart. After two three showers of heavy rains go for transplanting of plants. High yielding varieties like L49, Sardar, and Allahabad Safeda are recommended. In Guava fruit skirting bags may be used to cover fruits against sucking pest and sunburn injury. Avoid harvesting guavas during heavy rainfall, as the fruit may be damaged or its quality may be reduced. Store harvested guavas in a cool, dry place to prevent spoilage. During the flowering /fruiting stage, install

pheromone traps @ 18 - 20 per ha to control the fruit fly damage. For management of dieback spray zinc sulphate 2.0 kg + lime l.0 kg in 100 L of water and keep fruit fly trap to control fruitfly or spray Spinosad 0.4 ml per liter of water or lambda cyhalothrin 10WP 1.25 g per L of water. For management of Mealybug spray profenophos 1 ml + acephate 1 g + 10 g or soap powder Nematodes bio-control using nematode egg parasitic fungus Paecilomyceslilacinus @25 grams along with farm yard manure @ 100 kg, neem cake @ 250 g. For management of Fruitfly: Collect and destroy fallen and infested fruits summer ploughing to expose pupa. Use methyl eugenol lure trap (25/ha) to monitor and kill adults of fruit flies. Prepare methyl eugenol and Malathion 50 EC mixture at 1:1 ratio (take 10 ml mixture/ trap). Insecticides: Malathion 50 EC 0.05%. Bait spray combining molasses or jaggery 10g/l and one of the insecticides, Malathion 50 EC 2 ml/l, dimethoate 30 EC 1ml/L, two rounds at fortnight interval before ripening of fruits.

Grapes

Maintain a well-aerated canopy to reduce humidity within the vineyard and improve air circulation. Monitor for any nutrient deficiencies, particularly potassium, and apply appropriate fertilizers as needed. Keep the orchard clean by removing weeds and debris to prevent fungal and bacterial diseases. Ensure proper drainage and prevent waterlogging. The vigorous varieties of grapes the tendrils and the twins may be pruned. Management of April pruning in grape orchard should be done as per availability of water. During rest period, provide only need based irrigation to protect the existing leaves from drying, and also contribute towards increasing the reserves of the vines through photosynthetic activity. Care should be taken to reduce/stop the water in case new growth is observed on the shoot.

KHANDESH/ NORTHERN MAHARASHTRA

This region of Maharashtra is comprised of Jalgaon, Dhule, Nandurbar, Nashik). Rainfall of the region is 600-900 mm and it has Medium to deep black soils. This region has state agricultural university viz., Mahatma Phule Krishi Vidyapeeth, Rahuri, Ahmednagar (M.S)

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COMMERCIAL CROPS

Cotton

Use medium to deep black cotton soils with good drainage for optimal crop performance. Prepare land with 1-2 deep ploughings followed by harrowing; adopt ridge-furrow or broad bed-furrow method. Treat seeds with *Trichoderma viride* or Carbendazim for fungal diseases and imidacloprid for sucking pest protection. Sow *Bt* cotton hybrids like RCH-2 BG II or NCS-855 BG II from mid-June to early July, depending on monsoon onset. Maintain spacing of 120 × 60 cm for hybrids and 90 × 45 cm for varieties; use seed rate of 450-500 g/acre. Apply basal dose of NPK @ 50:25:25 kg/ha based on soil test; supplement with FYM @ 5-10 t/ha. Apply zinc sulphate @ 25 kg/ha in zinc-deficient soils. Ensure field has proper drainage to prevent waterlogging, especially during early growth stages. Provide protective irrigation during long dry spells, particularly during flowering and boll development. Apply Pendimethalin @ 1.0 kg a.i./ha as pre-emergence herbicide;

follow with hand weeding at 20-25 DAS. Monitor for sucking pests (jassids, aphids, whiteflies); control using imidacloprid 0.3 ml/l or thiamethoxam 0.25 g/l.

Further, install 5 pheromone traps per hectare to monitor bollworms. If needed spray emamectin benzoate or flubendiamide as per threshold levels. Manage leaf spots and blights with mancozeb or copper oxychloride sprays as needed. Apply top dressing of nitrogen at 30 and 60 DAS for vigorous growth. Use plant growth regulators like NAA to minimize flower and boll drop, if observed. Harvest mature, dry bolls; avoid harvesting during or immediately after rain to prevent boll rotting.

PULSE CROPS

Soybean

The crop must be sown before 1st week of July, after good rain, with a spacing of 45×5?cm; depth <4?cm. the seed must be treated with Carboxin + Thiram 30g/10kg + Thiamethoxam 100ml/10kg. Give a hoeing at 15-20 DAS. Apply water at branching (30-35 DAS) & flowering (40-45 DAS).

Pigeon pea

Recommended varieties for sowing include Phule Trupti, BDN-711, and Renuka, with planting done before the first week of July. Suggested spacing is 60×20 cm, 90×20 cm, or 180×30 cm, depending on field conditions. Treat seeds with Trichoderma (5 g/kg) or a combination of Carboxin and Thiram (4 g/kg), along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 25:50 kg N:P per hectare. The seed rate varies-12-15 kg/ha for normal spacing and 5-6 kg/ha for wider spacing. Intercropping is recommended with bajra, soybean, sorghum, cotton, moong, or urad for better land utilization. Irrigate at 30-35 days after sowing (DAS), 60-70 DAS, and during the pod-filling stage. Perform one hoeing at 15-20 DAS followed by hand weeding for effective weed control.

Black gram

The recommended seed rate is 15-20 kg/ha with a spacing of 30×10 cm. Seeds should be treated with Trichoderma at 5 g/kg along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 20:40 kg N:P per hectare. Intercropping with pigeon pea and other kharif crops is advised for better resource utilization. Irrigation should be provided at critical stages such as flowering and pod filling to ensure healthy growth and yield.

FRUIT & VEGETABLE CROPS

Pomegranate (Mrig Bahar)

Pomegranate cultivation during Mrig Bahar requires pruning to remove diseased branches, followed by an application of 1% Bordeaux mix. Pre-flowering fertilization includes 700 g urea, 1600 g SSP, and 400 g MOP per tree. Dusting with 4% copper dust or bleaching powder helps in disease prevention. Regular pest control with Azadirachtin or alternate insecticides every 15 days is crucial. At the fruiting stage, apply 500 g DAP + 100 g MOP per tree at the lemon size stage, and 200 g 19:19:19 + 100 g MOP at the guava size stage.

Grape

After pruning, grapevines are treated with Hydrogen Cyanamide (30-40 ml/L) on the top 2-3 buds to encourage growth. GA sprays of 10 ppm are applied 17-18 days after pruning, followed by 15 ppm after 4-5 days. Mealybug control in September involves removing bark and spraying Methomyl 40 SP at 1 g/L

Banana

Banana is planted on raised beds with a paired row system, ideally after 2-3 monsoon showers. Varieties like Grand 9 and Basarai are recommended. Green manuring and proper drainage are crucial during heavy rains. Preventive soil applications of Pseudomonas and Trichoderma help manage diseases. Practices like de-navelling during fruiting, irrigation at fruit development, and spraying Propiconazole for leaf spot management are recommended for healthy crop growth. During flowering/fruiting stage, carry out de-navelling (removal of male inflorescences for nutrient diversion) practices for better growth of banana fruits. Apply irrigation at fruit development stage. At the time of planting treat the sucker with Mancozeb (3g/l) and Monocrotophos @ 2ml/l, apply *Trichoderma viride* @ 5 kg) + FYM @ 100 kg for management of soil borne diseases. For management of Sigatoka leaf spot disease in banana orchard, spraying of 0.5 ml Propiconazole 25 EC + Mineral oil 1% per litre of water.

Onion (Kharif)

For effective nursery preparation, raise beds enriched with FYM, copper oxychloride, and a balanced mixed fertilizer. Prior to sowing, treat seeds with Thiram or Carbendazim at 2-3g/kg to prevent fungal infections. Apply Basalin at 25ml per 10 L of water as a pre-emergence herbicide for effective weed control. At planting, use a fertilizer dose of 100:50:50 NPK, with 50:50:50 applied at sowing and the remaining 50?N at 45 days after transplanting (DAT). To manage fungal diseases, integrate Trichoderma along with regular fungicide sprays every 10-15 days. For thrips control, install sticky traps and use Profenofos, Fipronil, or Spinosad as required based on pest observation.

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For effective nursery preparation, raise beds enriched with FYM, copper oxychloride, and a balanced mixed fertilizer. Prior to sowing, treat seeds with Thiram or Carbendazim at 2-3g/kg to prevent fungal infections. Apply Basalin at 25ml per 10 L of water as a pre-emergence herbicide for effective weed control. At planting, use a fertilizer dose of 100:50:50 NPK, with 50:50:50 applied at sowing and the remaining 50?N at 45 days after transplanting (DAT). To manage fungal diseases, integrate Trichoderma along with regular fungicide sprays every 10-15 days. For thrips control, install sticky traps and use Profenofos, Fipronil, or Spinosad as required based on pest observation.

KONKAN REGION

Konkan Region of Maharashtra is comprised of Mumbai, Mumbai Suburban, Thane, Palghar, Raigad, Ratnagiri, Sindhudurg). The region has high rainfall (3000-4000mm) and has Lateritic, acidic, high organic matter. This region possesses a state agricultural university viz., Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli.

CEREAL CROPS

Rice

Rice nursery sowing should begin with the onset of monsoon showers, typically in the first to the second week of June. It is crucial to select suitable varieties like Ratnagiri-1, Karjat-3, Karjat-4, and Sahyadri based on local climatic conditions and market demand. For one hectare of the main field, approximately 60-75 kg of seed is required, and a nursery area of around 1000 m^2 should be prepared. Soak rice seeds in water for 24 hours, followed by incubation for 24-36 hours. Treat the seeds with fungicides like Carbendazim (2g/kg seed) or biocontrol agents such as *Trichoderma*

to prevent seed-borne diseases. In regions with heavy rainfall, wet bed nurseries are recommended, especially on low-lying lateritic or clay loam soils. Where drainage is poor, raised beds should be prepared. Select well-drained and fertile fields near water sources, avoiding waterlogged or saline areas. Apply 2 kg of urea and 1 kg of SSP per 100 m^2 of nursery, and incorporate well-decomposed FYM or compost (5-10 t/ha) during bed preparation.

Pre-germinated seeds should be uniformly broadcasted on nursery beds with light pressing to ensure proper soil contact. Maintain a shallow water level of 2-3 cm during the early growth stage, avoiding stagnation to prevent diseases. Weed pressure is generally low, but light raking or manual weeding should be done 10-12 days after sowing if required. To manage weeds effectively, spray Butachlor 50 EC @30 ml/litre water 2-3 days after sowing. Transplant 21-day-old seedlings with a spacing of 20×15 cm for normal varieties and 15×15 cm for short-duration varieties. At transplanting, apply 10 t/ha of Gliricidia leaves or 8 t + 50 kg of nitrogen for better puddling and root establishment. Give the second nitrogen dose (1 kg urea/guntha) 15 days after transplanting to boost growth. Replant damaged or gap areas promptly with healthy seedlings to maintain uniformity

Fertilization and Pest/Disease Control: Apply 3 kg of compost per m^2 of nursery bed before sowing. For mechanized farming, mat nurseries on plastic sheets or trays are recommended, with fertilizers applied at 2.2 kg urea, 3 kg SSP, and 0.8 kg MOP per guntha. For effective pest control, monitor the nursery for stem borers and use Quinolphos (150 ml/guntha) or Carbofuran (165 g/guntha) as needed. Crab control can be managed using toxic bait prepared with Acephate or Carbaryl mixed with cooked rice. Monitor for diseases like blast, brown spot, and root rot, and apply fungicides and insecticides as per local recommendations.

MILLET CROPS

Finger Millet (Nagli) & Proso Millet (Vari)

For effective nursery preparation, sow the rice seeds in rows on 10 guntha nursery beds using a seed rate of 6 kg per hectare. Fifteen days after sowing, apply Quinolphos at 15 kg/ha or Carbofuran at 16.5 kg/ha to manage early pest infestations. To prevent blast disease, spray Zineb at 2 kg or Copper Oxychloride at 1.25 kg mixed in 500 liters of water per hectare. Transplant healthy 4-5-week-old seedlings at a spacing of 15×20 cm, and ensure proper nutrient application with a recommended dose of 40:40 $N:P_2O_5$ kg/ha to promote vigorous growth and better yield.

OILSEED CROPS

Groundnut

Incorporate 5-7.5 t/ha of well-decomposed FYM during the final tillage to enhance soil fertility. At the time of sowing, apply 25 kg of Nitrogen and 50 kg of Phosphorus per ha using boronated SSP for better root development. Use 100-110 kg of seed per hectare from recommended varieties like Konkan Bhuratna and TAG-24. For effective disease control, treat seeds with Thiram at 5 g/kg along with Rhizobium culture at 25 g/kg before sowing. Follow a sowing pattern of 30 cm paired rows with

OILSEED CROPS

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15 cm spacing between plants for optimal growth. Ensure proper weed management during the first 30-45 days to avoid competition and promote healthy crop establishment

FRUIT & VEGETABLE CROPS

Mango (New Planting)

For successful mango plantation, select well-drained sites and use raised beds if necessary. Prepare planting pits by filling them with a mixture of topsoil, compost, and 2 kg of SSP. Ensure the graft joint remains above the soil surface, and provide proper staking. Maintain orchard hygiene and consider 10-20% inter-varietal grafting for better cross-pollination. Before planting, dust pits with 100 g of Folidol 2% or Carbaryl 10% to prevent pests.

Use of Skirting bag in Mango

Cashew

Plant grafts from improved, disease-resistant varieties in pits enriched with compost and 1 kg of bone meal or SSP. When using polybag grafts, remove the bags carefully to avoid damaging the roots. After planting, remove any pegs or bands that might restrict plant growth for optimal development.

Coconut & Betel Nut

Maintain cleanliness in coconut and betel nut gardens by composting dry leaves and sticks. To control mildew in betel nut plants, apply a 1% Bordeaux mixture as a preventive spray.

For better yield, use improved seeds for gourds like padwal and dudhi, and train the vines on trellises for proper growth. For okra, select virus-resistant varieties such as Arka Anamika and ensure planting in well-drained soils. Brinjal and chilli should also be cultivated in well-drained areas with the application of 50 kg of nitrogen and 75 kg of phosphorus per hectare for optimal growth.

Pepper

For pepper cultivation, spray a 1% Bordeaux mixture on old vines to control fungal infections. Ensure proper drainage and provide adequate support for the climbing vines to promote healthy growth and prevent damage.

In animal husbandry, protect animals from stray dogs and rabies. Use rubber mats to improve cow comfort and enhance milk yield. Raise calves from high-milk-yielding cows and ensure the maintenance of genetically superior bulls. Rotate bulls every 3-4 years between villages to maintain genetic diversity.

For fisheries, in freshwater ponds, clean and remove debris after harvesting before restocking. In brackish water ponds, dry and apply lime post-harvest to prepare for the next shrimp culture cycle.